Commitment Can Be Bad

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1 Introduction

As Artificial Intelligence (AI) technology continues to evolve, AI agents are becoming an increasingly prominent topic. In the future, we may see capable AI agents operating under minimal oversight engaging in strategic interactions. It has been suggested that such AI agents could have the ability to make credible commitments (**hammond˙multi-agent˙2025**). There are concerns that if AI agents possess such an ability to make commitments, they might use this power for extortion. For example, one AI agent might commit to an aggressive action that then forces another AI agent to accommodate. But if both AI agents possess this capability, each may attempt to commit to the aggressive action first. This dynamic gives rise to what has been coined a commitment race (**kokotajlo˙commitment˙2019**), in which AI agents rush to commit before the other. If both commit simultaneously to incompatible strategies, the outcome could be harmful to both.

In this thesis, I develop a formal model of simultaneous commitment to study such dynamics. Two players play a simple game, referred to hereafter as the base game. Before playing the base game, players have the opportunity to simultaneously commit to any normal-form strategy of this game. Commitment is voluntary; players can choose not to commit and wait until the base game. When players play a base game with the option to commit, I refer to this as commitment game. My main result is that in commitment games equilibria can emerge that are worse in payoffs for all players than the corresponding equilibria in the base game. This arises because the non-aggressive strategy of waiting in the commitment game becomes less attractive

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than non-aggressive behavior in the base game, and thus aggressive behavior rises. I will illustrate this with an example.

Example

Consider the well-known Game of Chicken depicted in Table 1, where two players simultaneously decide whether to go Straight or Move (out of the way). Besides two pure-strategy Nash equilibria, this game has one mixed-strategy Nash equilibrium (MSNE), in which both players choose Straight with probability $\frac{2}{3}$ and Move with probability $\frac{1}{3}$, resulting in an equilibrium payoff of $-\frac{1}{3}$. Importantly, this MSNE is the unique symmetric equilibrium for this symmetric game.

	Straight	Move
Straight	$-1, -1$	$1, -0.5$
Move	$-0.5, 1$	$0, 0$

Table 1: A Simple Game of Chicken

How would equilibrium outcomes change if players could simultaneously commit to their strategies beforehand? Figure 1 presents a commitment variant of this game, with an initial commitment period. Initially, both players simultaneously decide either to commit to Straight (S) or Move (M), but they can also decide to wait (W). After this commitment period, the made commitments are observed and the base game is played. If both players commit, their chosen strategies are executed, concluding the game. If only one player commits, the other observes this commitment and selects their strategy accordingly. Should both players choose to wait, they proceed to play the base Game of Chicken.

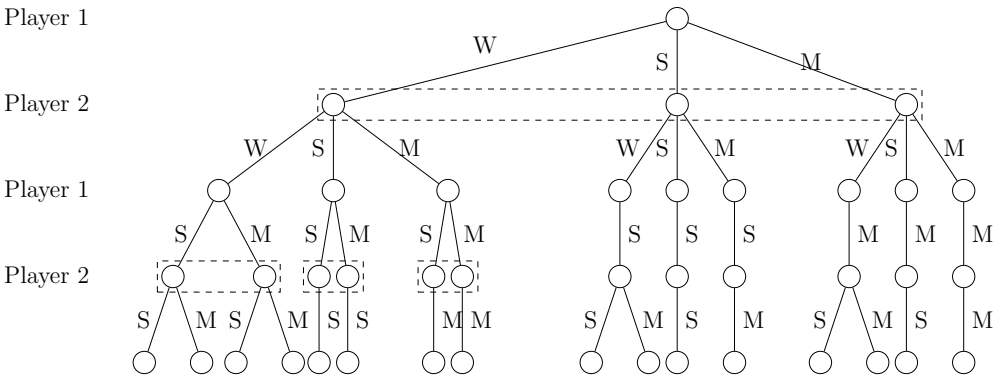


Figure 1: The Game Tree for the Commitment Game of Chicken

In this commitment game, there are multiple subgame-perfect Nash equilibria (SPNE) in pure strategies: in one, a player commits straight while the other waits and then

moves out of the way; in another, one player commits straight and the other commits to moving out of the way. Importantly, no equilibrium can involve both players waiting before choosing a pure-strategy equilibrium, since in that case one player would end up in her less-preferred equilibrium and could profitably deviate by committing straight. Next to these pure strategy equilibria, there exists an interesting mixed subgame perfect equilibrium with probabilistic commitment. Here, each player commits to the aggressive strategy (Straight) with probability $\frac{8}{11}$ and waits with probability $\frac{3}{11}$. If both players opt to wait, they play the base game and the original MSNE strategies there. This mixed equilibrium is the unique equilibrium with probabilistic commitment, and it is also the unique symmetric equilibrium. Strikingly, this equilibrium has expected payoffs of $-\frac{5}{11}$ per player, representing a lower expected payoff than the original game's mixed-strategy equilibrium with $-\frac{1}{3}$.[1] The reason is that players are more likely to behave aggressively. While players move with probability $\frac{1}{3}$ in the base game's MSNE, they only wait with probability $\frac{3}{11}$ in the commitment game. Introducing commitment leads to more aggressive behavior, increased miscoordination, and lower payoffs for both players.

Literature

In economic theory, commitment is often associated with welfare-enhancing outcomes, at least for the player making the commitment. Commitment features prominently across a wide range of models, such as principals committing to mechanisms, players delegating decision-making to agents, and first movers in Stackelberg games earning a higher profit (**kalai˙commitment˙2010**). This thesis, however, turns to the question of whether the ability to make commitments can *harm* players.

Existing economic theory literature has largely focused on one-sided commitment, leaving two-sided commitment relatively understudied. **harrenstein˙commitment˙2007** analyze two- and three-player games in which all players have the ability to commit. However, the authors assume a predetermined order of commitment and do not allow for simultaneous commitment.

kalai˙commitment˙2010 present a model of simultaneous conditional commitment, expanding on earlier work by **tennenholtz˙program˙2004**. In their framework, players select commitment devices d before the base game. A commitment device specifies a base game strategy as a function of the other's chosen commitment device, $d_i(d_{-i}) = S_i$. The authors' main result is a commitment folk theorem,

[1] For clarity, in this thesis, only equilibria involving both-sided probabilistic commitment are referred to as mixed equilibria in commitment games, even though pure strategy equilibria are technically also mixed strategy equilibria.

stating that the Nash equilibria in such commitment games span all individually-rational correlated strategy payoffs of the game. Naturally, this result relies on the availability of a complete space of conditional commitment devices. In contrast, my model limits players to committing exclusively to normal form strategies. This still allows for some conditionality in sequential base games. For example in an entrant-incumbent game, the incumbent's normal form strategies allow commitment that conditions on the entrant's initial decision whether to enter the market or not.

The remainder of this thesis is organized as follows. Section 2 shows that, in addition to cases where payoffs decline, there are also base games in which commitment raises payoffs for both players. Here, Proposition 1 characterizes symmetric 2×2 games in which commitment is either beneficial or detrimental. Section 3 considers games with multiple commitment periods T. Here I present Proposition 2, which establishes a convergence result as $T \to \infty$. I also introduce a game where commitment leads to mutual extortion. Section 4 considers an extension where, instead of simultaneous commitment, there is uncertainty over the order of commitment. Section 5 briefly considers commitment outside of 2×2 games. Section 6 concludes.

2 Commitment in 2×2 Games

In the introduction, I presented a game in which commitment leads to strictly lower equilibrium payoffs. In this section I will investigate for which broader class of games this result holds. I analyze the effect of introducing commitment on equilibrium payoffs in a class of symmetric 2×2 games. Specifically, I will consider 2×2 games that are symmetric in the sense of table 2 (left) with $A < D$. To reduce the number of parameters without loss of generality, I apply a positive affine transformation: subtracting A from all payoffs and dividing by $D - A$. This allows me to redefine the remaining parameters as $b = \frac{B-A}{D-A}$ and $c = \frac{C-A}{D-A}$, resulting in the normalized form shown in Table 2 (right). This transformation follows standard practice (**tewolde˙game˙2024**).

	A_1	A_2
A_1	(A, A)	(B, C)
A_2	(C, B)	(D, D)

$\longrightarrow$

	A_1	A_2
A_1	$(0, 0)$	(b, c)
A_2	(c, b)	$(1, 1)$

Table 2: Representation of a general 2×2 game (left) and its simplified form (right).

Equilibria without Commitment

Different values of b and c represent different types of well known 2×2 games. Figure 2 illustrates this. When $b > 1$ and $c \in [0, 1]$, the game takes the form of a Game of

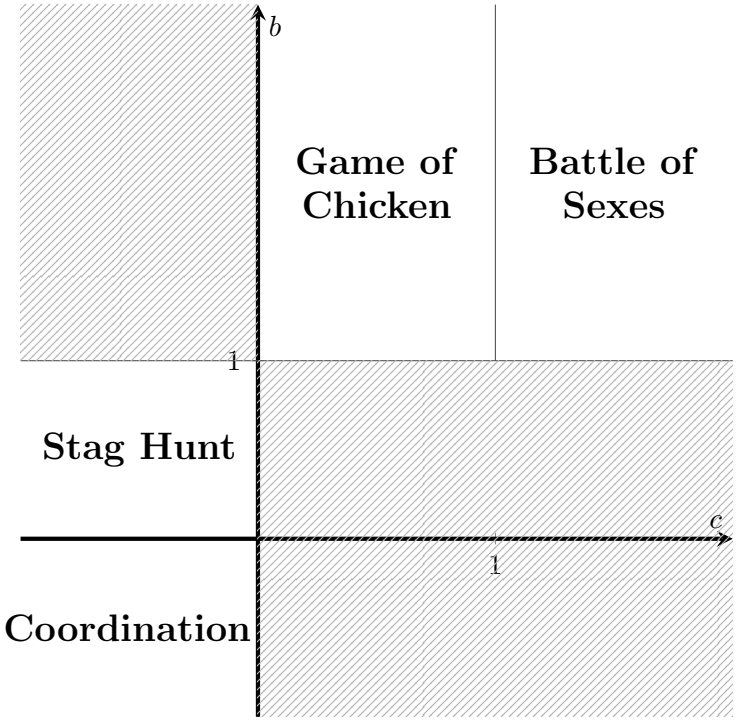


Figure 2: Game-type regions in (c, b)-space.

Chicken. Games with $b > 1$ and $c > 1$ are known as the Battle of Sexes. The Stag Hunt game arises when $b \in [0, 1]$ and $c < 0$. Finally, games with $b < 0$ and $c < 0$ do not really have a standard game name; I refer to this residual class as Coordination games. Shaded areas denote games with strictly dominated strategies.

In pure strategies, the Game of Chicken and the Battle of Sexes have the Nash equilibria $\{A_1, A_2\}$ and $\{A_2, A_1\}$, whereas Stag Hunt and Coordination have the equilibria $\{A_1, A_1\}$ and $\{A_2, A_2\}$. All of these games have one equilibrium that involves mixing: Players play A_1 with probability $\frac{b-1}{c+b-1}$, which yields the expected payoff $\frac{bc}{b+c-1}$. [2] Note that this MSNE is the unique symmetric equilibrium for the Game of Chicken and the Battle of Sexes.

Equilibria with Commitment

Base game equilibria in pure strategies translate into subgame perfect Nash equilibria in pure strategies in the commitment game. This can occur in three distinct ways:

[2] The general game does not have a mixed equilibrium when $b + c = 1$, in which case there are (weakly) dominated strategies.

both players committing upfront to base game equilibrium strategies, one player committing first and the other waiting and then best responding, or both players waiting and then subsequently playing equilibrium strategies. In the appendix in section A, I establish two lemmas characterizing the conditions under which each type of pure strategy equilibrium arises.

However, I argue that focusing on subgame perfect equilibria in *mixed strategies* rather than pure strategies is considerably more compelling for two reasons: symmetry and equilibrium selection.

Symmetry All 2×2 games considered in this section are symmetric, yet the Game of Chicken and the Battle of Sexes do not have a symmetric equilibrium in pure strategies, neither in the base game nor in the commitment version. Asymmetric equilibria seem counterintuitive. Players are identical in both strategy sets and payoffs, thus there is no clear rationale for why one should expect opposite equilibrium behavior. As I show, the mixed equilibrium is the unique symmetric equilibrium in these games, and thus seems better fitted as a solution concept to predict the behavior of players.

Equilibrium Selection In each commitment game, multiple pure strategy Nash equilibria typically exist. Given the absence of communication devices and the one-shot nature of commitment games, it seems implausible that players can seamlessly coordinate on a single equilibrium. This is particularly relevant in the Game of Chicken and the Battle of Sexes, where each player prefers a different equilibrium. A mixed equilibrium naturally incorporates the possibility of miscoordination, quantifies the probability with which miscoordination might occur, and provides expected payoffs rather than deterministic outcomes. Additionally, Harsanyi's purification theorem (**harsanyi˙games˙1973**) further supports the practical relevance of mixed equilibria.

Proposition 1
Consider a symmetric 2×2 *game and its commitment variant. In the unique mixed subgame perfect equilibrium of the commitment game, the expected payoff for both players relative to the mixed equilibrium of the base game is:*

1. *weakly lower in the Game of Chicken (strictly for* $c < 1$*);*

2. *higher in the Battle of Sexes.*

Notably, Proposition 1 establishes that commitment can induce equilibria in which both players are strictly worse off. To my knowledge, this finding is not documented

in the existing literature. The proof is presented in the appendix in section B, where the equilibria for all game-type regions are derived.

Proposition 1 illustrates that the impact of commitment on player payoffs depends on the underlying game. In the Game of Chicken, commitment makes players worse off; in contrast, it improves outcomes in the Battle of the Sexes. To see why, consider the Game of Chicken with commitment. Committing to A_2 (moving out of the way) is weakly dominated by waiting, so players effectively choose between committing A_1 and W. The resulting reduced normal form is shown in Table 3, which closely resembles the generalized 2×2 base game in Table 2, except that A_2 is replaced by W, and the (A_2, A_2) payoff is substituted by the "Wait–Wait payoff." In the base game, if both players play the non-aggressive strategy A_2, they receive the safe payoff 1. However in the commitment game, if both play the non-aggressive strategy W, they receive the expected mixed equilibrium payoff of the base game, $\frac{bc}{c+b-1}$. This payoff is less than 1 in the Game of Chicken, where $b > 1$ and $c \in (0, 1]$. As a result, commitment transforms the Game of Chicken into one with worse outcomes. In the Battle of the Sexes, this dynamic is reversed. Since $b > 1$ and $c > 1$, the Wait-Wait payoff $\frac{bc}{c+b-1}$ exceeds 1, so commitment improves payoffs relative to the base game.

	A_1	W
A_1	$0, 0$	b, c
W	c, b	$\frac{bc}{c+b-1}, \frac{bc}{c+b-1}$

Table 3: Reduced Normal Form of the Game of Chicken with Commitment

While initially stated that all symmetric 2×2 games would be considered, Proposition 1 focuses only on the Game of Chicken and the Battle of the Sexes. Corollary 1, derived in Appendix B, extends the analysis to the Stag Hunt and Coordination Game.

Corollary 1

The following results hold for the Coordination Game and the Stag Hunt.

1. *There is a mixed subgame perfect equilibrium in the Coordination Game. However, players committing to (A_1, A_1) is another symmetric subgame perfect equilibrium that Pareto-dominates the mixed equilibrium.*

2. *The Stag Hunt game does not have a mixed equilibrium.*

Although Corollary 1 is instructive, the more interesting strategic dynamics arise in games such as the Game of Chicken and the Battle of the Sexes. Thus, in the next section, I will focus on these two game types.

3 Multiple Commitment Periods

So far, I have assumed a single commitment period before the base game, during which players can simultaneously commit. In this section, I extend the model to allow for multiple commitment periods, denoted by T. If players can credibly commit to a strategy for a game they will play in the future, it seems natural that they might be able to do so at several points in time leading up to the game. This is particularly relevant if one thinks of these players as AI agents. Fast computational speed could imply that agents have many, even effectively infinite, opportunities to commit.

From this point forward, let T represent the number of discrete commitment periods. If one player commits in the first commitment period, this commitment is irreversible and the other player will best respond to it. Should both players not commit in the first period, they observe each other's inaction and then in the second period simultaneously decide again whether to commit. The base game is played only if neither player commits in any of the T commitment periods.

Let A_{1t} denote the strategy whereby a player waits until period t to commit to A_1, unless the other player has already committed in an earlier period. In that case, the player best responds. Waiting (W) is the strategy of not committing in any commitment period, best-responding to any commitment made by the other player, and if the base game is reached playing the MSNE strategy of the base game. Let p_t denote the probability of a player playing A_{1t} and let q_t denote the probability of a player playing A_{2t}. The probability of waiting is denoted as $\sigma(W) = 1 - \sum_{\tau=1}^{T}(p_\tau + q_\tau)$.

Proposition 2

As $T \to \infty$, the expected payoff in the unique mixed subgame perfect equilibrium for both players

1. *decreases in the Game of Chicken and converges to c*
2. *increases in the Battle of Sexes and converges to* $\min\{b, c\}$

The full proof of proposition 2 can be found in the appendix in section C. I prove three lemmas to establish that this is indeed the unique mixed subgame perfect equilibrium for all T, and to characterize the equilibrium as $T \to \infty$, including closed-form expressions for all p_t, q_t, and $\sigma(W)$ in terms of b, c, and T. Table 4 describes the probabilities of committing in the first period and waiting as $T \to \infty$, as well as the equilibrium payoffs.

To illustrate Proposition 2, recall the example commitment Game of Chicken from the introduction. With a single commitment period, $T = 1$, the mixed-strategy

Condition	Asymptotic Behavior	Payoff
$0 < c \leq 1 < b$	$p_1 \to \frac{b-c}{b}, \quad \sigma(W) \to 0$	$\to c$
$1 < c < b$	$p_1 \to \frac{b-c}{b}, \quad q_1 \to 0, \quad \sigma(W) \to 0$	$\to c$
$1 < b < c$	$p_1 \to 0, \quad q_1 \to \frac{c-b}{c-1}, \quad \sigma(W) \to 0$	$\to b$

Table 4: Asymptotic convergence of probabilities as $T \to \infty$.

equilibrium involves both players committing to Straight with probability $\frac{8}{11}$, waiting with probability $\frac{3}{11}$, and receiving an expected payoff of $-\frac{5}{11}$. When the number of commitment periods increases to $T = 2$, the mixed equilibrium changes: each player commits to Straight in the first period with probability $\frac{96}{129}$, in the second period with probability $\frac{24}{129}$ (conditional on no one having committed in the first period), and waits in both periods with probability $\frac{9}{129}$. The resulting expected payoff is $-\frac{21}{43}$, which is strictly lower than under $T = 1$. As T increases further, the mixed equilibrium payoff further decreases. The probability of committing Straight in the first period increases and converges to $\frac{3}{4}$. The expected payoff converges to $-\frac{1}{2}$ for both players, which is the payoff of 'losing' the Game of Chicken.

This convergence occurs because each additional commitment period hurts both players. As previously argued, introducing one period of commitment transforms the base Game of Chicken into a game with lower payoffs, as illustrated in Table 3. Adding another commitment period to this game transforms it into a game with even lower payoffs. In the first commitment period, players choose whether to commit Straight or wait. If both wait, the game proceeds. Under $T = 1$, this would lead to the MSNE of the base game. Under $T = 2$, however, it leads to the $T = 1$ commitment game, which has a lower expected payoff. As a result, the equilibrium payoff is lower when $T = 2$. Each additional commitment period worsens outcomes in the Game of Chicken. As $T \to \infty$, both players effectively lose the Game of Chicken—no one wins.[3]

The dynamic described above can be characterized as a *commitment race*: players have strong incentives to commit early, leading to frequent miscoordination and low expected payoffs. Of course, both players committing to Straight in the first period is not an equilibrium, but in the mixed-strategy equilibrium this outcome occurs with probability $(\frac{3}{4})^2 \approx 56\%$ in the Game of Chicken as $T \to \infty$. The term commitment race was originally introduced by **kokotajlo˙commitment˙2019** in the context of

[3]However, for similar reasons commitment improves the Battle of Sexes.

AI agents potentially extorting one another. While his treatment is highly informal, my contribution here is to formalize this idea within a game-theoretic framework. To explore the implications for extortion, I now introduce a new game.

Extortion Game

Consider two players, each with a monetary endowment M. They simultaneously decide whether to keep their endowment or yield it to the other player. After observing each other's choices, each player can punish a refusal to yield by inflicting harm h on the other at a personal cost $c(h) > 0$, where $c(h)$ increases with h. Importantly, punishment decisions can depend on the other player's action.

Without commitment, threats of punishment are not credible because punishment involves a costly action decided upon only after the other player decided to keep or yield her endowment. Consequently, in the subgame-perfect equilibrium without commitment, both players keep their endowment and choose not to punish. With commitment, however, players can make credible threats. Players can commit in advance to punishing the other player if they refuse to yield. If punishment is severe enough ($h \geq M$), yielding becomes the best response. For simplicity, assume committing to a punishment level $h = M$ ensures yielding.

Table 5 shows the game's normal form with one period of commitment. Players can commit to all normal form strategies of the game, which consist of a choice between yielding or keeping, combined with a decision to punish or not punish. Players can also wait, in which case they best respond to any made commitment. If both wait, they play the subgame-perfect equilibrium of the base game.

	Yield, Punish	Yield	Keep, Punish	Keep	W
Yield, Punish	(M, M)	(M, M)	$(-c,\ M)$	$(-c,\ M)$	(M, M)
Yield	(M, M)	(M, M)	$(0,\ 2M)$	$(0,\ 2M)$	$(0,\ 2M)$
Keep, Punish	$(M,\ -c)$	$(2M,\ 0)$	$(-c,\ -c)$	$(M - c,\ 0)$	$(2M,\ 0)$
Keep	$(M,\ -c)$	$(2M,\ 0)$	$(0,\ M - c)$	(M, M)	(M, M)
W	(M, M)	$(2M, 0)$	$(0,\ 2M)$	(M, M)	(M, M)

Table 5: Extortion Game with $h = M$ and punishment cost c.

Note that in Table 5, most strategies are weakly dominated by waiting[4], and will thus not feature in the mixed subgame perfect equilibrium. The game simplifies to the reduced form shown in Table 6. This simplified structure corresponds to a Game of Chicken, allowing direct application of results derived in this and the previous Section. It follows from Proposition 1 that commitment makes the mixed equilibrium

[4]Notably, Keep is also weakly dominated by waiting, because I assume that a punishment of $h = M$ is enough for a player to prefer yielding to keeping.

payoff of this game worse. Additionally, Proposition 2 shows that as the number of commitment periods T grows, the probability that both players commit to *(Keep, Punish)* in the first period converges to $\frac{2M}{2M+c}$, and the expected equilibrium payoff approaches 0.

	Keep, Punish	W
Keep, Punish	$(-c,\ -c)$	$(2M,\ 0)$
W	$(0,\ 2M)$	(M, M)

Table 6: Reduced form Extortion Game

Thus, in the resulting mixed-strategy equilibrium, players become increasingly likely to adopt incompatible extortionate commitments so that each commits to keeping her endowment and punishing the other for doing the same. As $T \to \infty$, the expected payoff from this mutual extortion scenario collapses to the point where both players effectively lose their entire endowment. This demonstrates how commitment can give rise to extortion and severely inefficient outcomes.

4 Incomplete Information

So far, I have examined commitment games in which both players commit simultaneously. I now relax this assumption by introducing a game of incomplete information, where there is uncertainty over the order of commitment. At the outset, Nature draws one of three possible states:

- ω_i, player i commits first, with probability μ;
- ω_{-i}, player $-i$ commits first, with probability μ;
- $\omega_\emptyset$, both players commit simultaneously, with probability $1 - 2\mu$,

where $\mu \leq 0.5$. Note that when a player commits first in states ω_i and ω_{-i}, the second-moving player still has the opportunity to commit before the base game is played, should the first mover choose to wait. I model uncertainty as follows. When the state is ω_{-i}, player i moves secondand observes the state perfectly. This is because a player moving second must observe the other's prior commitment in order to best respond, and thus cannot face any uncertainty. However, when player i is called to act and observes no prior commitment, she cannot distinguish whether the true state is ω_i or $\omega_\emptyset$. Accordingly, I model the signal structure as follows:

$$\sigma_i(\omega_i) = \sigma_i(\omega_\emptyset) = t_\emptyset \quad \text{and} \quad \sigma_i(\omega_{-i}) = t_{-i}.$$

It follows that player i's beliefs conditional on observing $t_\emptyset$ are given by $P_i(\omega_i|t_\emptyset) = \frac{\mu}{1-\mu}$ and $P_i(\omega_\emptyset|t_\emptyset) = \frac{1-2\mu}{1-\mu}$, while $P_i(\omega_{-i}|t_{-i}) = 1$.

Equilibria

I will now analyze the equilibria under this information structure using the stylized Game of Chicken from the introduction as the base game (see Table 1).

When player i receives the signal t_{-i} (indicating player i moves second), her equilibrium strategy is straightforward. She best responds to any commitment observed from the first mover. However, if she observes the first mover choosing to wait instead of committing, she will commit to going straight (S) to ensure her preferred equilibrium outcome. Now consider a player i who receives the signal $t_\emptyset$. In this case, player i has three possible strategies: commit to going straight (S_1), commit to moving out of the way (M_1), or wait (W). Assume player $-i$, when also receiving $t_\emptyset$, randomizes among these strategies with probabilities p, q, and $1-p-q$, respectively. Then the payoffs for player i's pure strategies are:

$$
\begin{aligned}
U_i(S_1 \mid t_\emptyset) &= P_i(\omega_i \mid t_\emptyset) \cdot 1 + P_i(\omega_\emptyset \mid t_\emptyset) \cdot \Big[p \cdot (-1) + q \cdot 1 + (1-p-q) \cdot 1 \Big] \\
U_i(M_1 \mid t_\emptyset) &= P_i(\omega_i \mid t_\emptyset) \cdot \left(-\tfrac{1}{2}\right) + P_i(\omega_\emptyset \mid t_\emptyset) \cdot \Big[p \cdot \left(-\tfrac{1}{2}\right) + q \cdot 0 + (1-p-q) \cdot \left(-\tfrac{1}{2}\right) \Big] \\
U_i(W \mid t_\emptyset) &= P_i(\omega_i \mid t_\emptyset) \cdot \left(-\tfrac{1}{2}\right) + P_i(\omega_\emptyset \mid t_\emptyset) \cdot \Big[p \cdot \left(-\tfrac{1}{2}\right) + q \cdot 1 + (1-p-q) \cdot \left(-\tfrac{1}{3}\right) \Big].
\end{aligned}
$$

Recall that S_1 denotes the strategy of committing to Straight upon receiving $t_\emptyset$, while M_1 denotes the strategy of committing to Move. Note that M_1 is weakly dominated by W.

Symmetric Pure Strategy Equilibrium There is an equilibrium where both players commit to going straight after receiving $t_\emptyset$. Note that this corresponds to setting $p = 1$ in the pure strategy payoffs above. Then:

$$
U_i(S_1 \mid t_\emptyset) = \frac{3\mu - 1}{1-\mu} \quad \text{and} \quad U_i(W \mid t_\emptyset) = -\frac{1}{2}, \tag{1}
$$

and $U_i(S_1 \mid t_\emptyset) \geq U_i(W \mid t_\emptyset) \Leftrightarrow \mu \geq 0.2$. Thus, the symmetric subgame perfect Nash equilibrium with both players committing straight upon receiving $t_\emptyset$ exists when $\mu \geq 0.2$. The expected equilibrium payoff is $U = P(t_\emptyset) \cdot \frac{3\mu-1}{1-\mu} + P(t_{-i}) \cdot (-\frac{1}{2})$, which equals $2.5\mu - 1$ since $P(t_\emptyset) = 1 - \mu$ and $P(t_{-i}) = \mu$.

It is easy to see that there is no equilibrium in which both players wait after receiving $t_\emptyset$. A strategy of always waiting corresponds to $1 - p - q = 1$, but in that case $U_i(S_1 \mid t_\emptyset) > U_i(W \mid t_\emptyset)$, so waiting is not a best response to waiting.

Asymmetric Pure Strategies Equilibria There are also equilibria where one player commits to going straight after receiving $t_\emptyset$ and the other one waits. Let player $-i$ be the one who commits straight. Then player i's payoffs are as in Equation 1, and waiting is a best response when $\mu \leq 0.2$. From player $-i$'s perspective, $U_{-i}(S_1 \mid t_\emptyset) > U_{-i}(W \mid t_\emptyset)$ always holds when the other player waits ($1-p-q=1$), so she indeed prefers to commit straight. Hence, this is an equilibrium. Moreover, instead of waiting, player i could also choose to commit to moving out of the way (M_1) in equilibrium. These asymmetric pure strategy equilibria yield a payoff of 1 to the player committing straight, and $-\frac{1}{2}$ to the player who waits (or moves aside).

Symmetric Mixed Equilibrium There is also a symmetric mixed equilibrium where players mix over S_1 and W when receiving $t_\emptyset$. M_1 is weakly dominated by waiting and will not feature in a mixed equilibrium, thus $q=0$. The p that equalizes $U_i(S_1 \mid t_\emptyset)$ and $U_i(W \mid t_\emptyset)$ is $p = \frac{8-7\mu}{11(1-2\mu)}$, which lies in $[0,1]$ for $\mu \leq 0.2$. The expected equilibrium payoff is

$$U = P(t_\emptyset) \cdot U_i(S_1 \mid t_\emptyset) + P(t_{-i}) \cdot \left[p \cdot \left(-\tfrac{1}{2}\right) + (1-p) \cdot 1 \right], \tag{2}$$

which can be rearranged to $\frac{3\mu-5}{11} + \frac{-2\mu-23\mu^2}{22(1-2\mu)}$. Let us briefly zoom in on the dynamics in Equation 2. When the state is ω_i or $\omega_\emptyset$, player i receives $t_\emptyset$ and mixes as described above. She then receives the expected payoff $U_i(S_1 \mid t_\emptyset)$ or $U_i(W \mid t_\emptyset)$, both are equal by the definition of mixed equilibria. When the state is ω_{-i}, only $-i$ receives $t_\emptyset$ and mixes, while i receives t_{-i}, knows that she can commit only as the second mover and observes the outcome of $-i$'s mixing. With probability p, that outcome is S_1, and i as the second mover will move aside. However, with probability $1-p$, the first mover waits, in which case i will commit to going straight—inducing her preferred outcome.

Summary and Interpretation All symmetric equilibria are summarized in Table 7.

Condition	Strategy Profile	Expected Payoff
$\mu \leq 0.2$	$\forall i:\ \sigma_i(S_1 \mid t_\emptyset) = p,\ \sigma_i(W \mid t_\emptyset) = 1-p,$ where $p = \frac{8-7\mu}{11(1-2\mu)}$	$\frac{3\mu-5}{11} + \frac{-2\mu-23\mu^2}{22(1-2\mu)}$
$\mu \geq 0.2$	$\forall i:\ s_i(t_\emptyset) = S_1$	$2.5\mu - 1$

Table 7: Symmetric equilibrium strategies upon receiving $t_\emptyset$

Intuitively, when commitments occur simultaneously ($\mu = 0$), each player commits to going straight with probability $\frac{8}{11}$ and obtains an expected payoff of $-\frac{5}{11}$, matching the results from the Game of Chicken presented in the introduction. What happens as μ increases? Players gain more information about the order of commitments, which one might initially expect to reduce miscoordination and thus improve payoffs. However, this is not necessarily the case. For certain intermediate values of μ, equilibrium payoffs become strictly lower than under simultaneous commitment, as illustrated in Figure 3. Here, the dashed horizontal line represents the payoff under simultaneous commitment, while the solid line depicts equilibrium payoffs for the symmetric equilibrium across values of μ. Notably, equilibrium payoffs dip below the simultaneous commitment payoff for values approximately in the interval $[0.11, 0.22]$. Figure 4 further clarifies that in this interval, the probability of miscoordination is higher than under simultaneous commitment. [5] Hence, more information about the order of commitment can lead to greater miscoordination and worse outcomes.

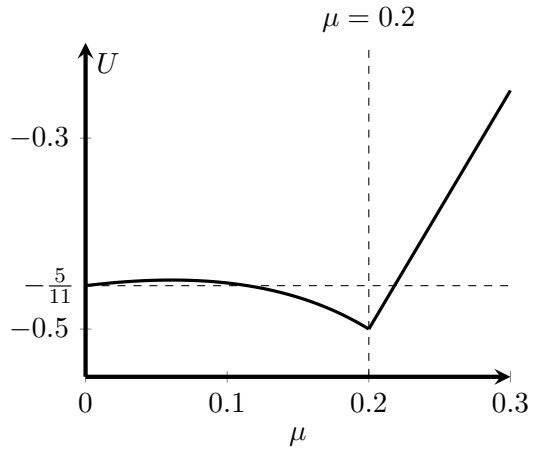


Figure 3: Equilibrium payoff

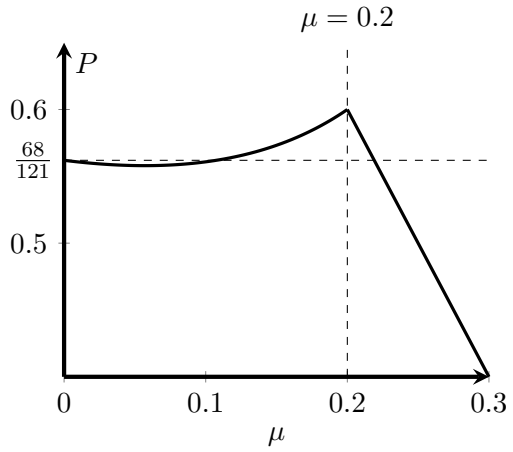


Figure 4: Miscoordination probability

To build intuition for this result, suppose player i decides to wait. If $\mu = 0$, the state is $\omega_\emptyset$ and two outcomes are possible: either the other player simultaneously committed, causing i to lose the Game of Chicken, or the other player also waited, leading to both playing the base game. When $\mu > 0$, a third outcome is possible: the state is ω_i, meaning i was actually the first mover. By waiting, i allows the second player to observe her inaction and immediately respond by committing straight, causing i to lose again. This makes waiting less attractive, prompting players to commit Straight more frequently. Consequently, the probability of miscoordination increases, lowering equilibrium payoffs. As μ grows larger, waiting eventually becomes dominated,

[5] Calculated as the sum of probabilities for miscoordinating directly on commitments and miscoordinating in the base game: $P(\omega_\emptyset) \cdot [p(S_1)^2 + (1 - p(S_1))^2 \cdot (\frac{2}{3})^2]$.

leading to both committing straight and achieving the lowest possible payoff. However, as μ continues to increase, simultaneous commitments become rarer, and the probability of miscoordination decreases again.

This concludes the current section. I have employed a stylized Game of Chicken to illustrate the dynamics arising under uncertainty about the order of commitment. Extending this approach to a wider class of games is an interesting future direction but falls outside the scope of this thesis.

5 Beyond 2×2

In this section I briefly explore the dynamics of commitment beyond 2×2 games. Consider the game in table 8. This game has the unique Nash equilibrium $\{A_1, A_1\}$. Furthermore note that for player 1, A_2 is strictly dominated by A_1, and for player 2, A_3 is strictly dominated by A_1. When commitment is introduced, there are two pure

	A_1	A_2	A_3
A_1	$1, 1$	$3, 0$	$\epsilon, 0$
A_2	$0, \epsilon$	$2, 1$	$0, 0$
A_3	$0, 3$	$0, 0$	$1, 2$

Table 8: A 3×3 game

strategy subgame perfect Nash equilibria: 1 commits to A_2 and 2 best responds with A_2, or 2 commits to A_3 and 1 best responds with A_3. Players now choose to commit to strategies that are strictly dominated in the base game (**harrenstein˙commitment˙2007**). Furthermore, there is an interior mixed equilibrium, where player 1 commits to A_2 with probability $\frac{1}{2}$ and waits with probability $\frac{1}{2}$ and player 2 commits to A_3 with probability $\frac{1}{2}$ and waits with probability $\frac{1}{2}$. Again, players make probabilistic commitments to strictly dominated strategies. The resulting expected payoff of this interior mixed equilibrium is 1.

Next, modify only the payoffs that the *opponent* receives when A_2 or A_3 is played, as in Table 9. Choosing A_2 still gives player 1 the same payoffs, but now imposes a large cost on player 2, and vice versa for A_3. The unique base-game equilibrium $\{A_1, A_1\}$ is unchanged, still giving each player a payoff of 1. With commitment, the game's mixed equilibrium becomes interesting: players still mix equally between committing A_2 and W, or committing A_3 and W, respectively. The expected payoff in this case drops drastically to $1 - \frac{10^9}{2}$ for each player, another demonstration of how commitment can be mutually harmful.

The games above illustrate that, in contrast to 2×2 games, 3×3 games can have

	A_1	A_2	A_3
A_1	$1, 1$	$3, 0$	$\epsilon - 10^9, 0$
A_2	$0, \epsilon - 10^9$	$2, 1 - 10^9$	$0 - 10^9, 0 - 10^9$
A_3	$0, 3$	$0, 0$	$1 - 10^9, 2$

Table 9: A modified 3×3 game

mixed equilibria with probabilistic commitment even if the base game only has one Nash equilibrium. Furthermore, while in the 2×2 case payoffs could decrease relative to the base game equilibrium, the difference in payoffs could not become too large, since both were functions of the same payoff parameters. By contrast, in the 3×3 case, this difference is less restricted and can grow arbitrarily large, as demonstrated. Overall, this section illustrated just one possible dynamic beyond 2×2 games. A more systematic exploration of such games would likely reveal further interesting phenomena, but lies beyond the scope of this thesis.

6 Conclusion

This thesis develops a model of simultaneous commitment, focusing on the possibility that commitment can lead to worse outcomes. In the model, players can voluntarily commit to normal-form strategies before playing a base game. I show that in some symmetric 2×2 games, the option to commit strictly reduces equilibrium payoffs. I characterize symmetric 2×2 games for which commitment reduces or improves equilibrium payoffs and extend the model to multiple commitment periods. As $T \to \infty$, early incompatible commitments can become more likely, and payoffs can decline further. I introduce an Extortion Game to illustrate how commitment can be used to make threats, and show that partial information about commitment order can increase miscoordination. Finally, I illustrate commitment in a stylized 3×3 game.

Several limitations remain. First, the finding that commitment can worsen outcomes relies on mixed-strategy Nash equilibria as the equilibrium concept, except in the incomplete information setting discussed in Section 4, where for some values of μ, pure strategy equilibria also lead to inefficient outcomes. I focus on mixed-strategy equilibria because they capture the possibility of miscoordination and allow for quantifying its likelihood. Moreover, in symmetric games, mixed equilibria may be the only symmetric equilibria available. However, this justification does not extend to asymmetric games.

Secondly, this thesis considers only a narrow class of 2×2 games. Analyzing a broader class of games may reveal additional dynamics. In particular, it would be

interesting to study two-sided simultaneous commitment in Cournot competition, given that one-sided commitment corresponds to the well-known Stackelberg model.

Third, in the setting with uncertainty over the order of commitment, I only consider a stylized Game of Chicken and a single commitment period. Future work could extend this to a more general class of games, multiple commitment periods, or asymmetries in the uncertainty over the commitment order.

Finally, while motivated by concerns over AI agents capable of making commitments, this thesis does not explore technical implementation. An interesting direction for further research is to explore in what ways commitment in AI agents can be designed to avoid the inefficient outcomes identified in this thesis. Conditional commitments (**kalai˙commitment˙2010**) or simulation-based program equilibria (**cooper˙characterising˙2024**) seem like promising directions here.

Appendix

A Commitment Game Equilibria in Pure Strategies

In Section 2, I noted that pure strategy equilibria of the base game carry over to the commitment game. In this section, I formally establish this connection by proving two supporting lemmas. Lemma 1 shows that any pure strategy equilibrium of the base game, such as (Straight, Straight) in the Game of Chicken, also constitutes an equilibrium outcome in the commitment game. In contrast, Lemma 2 demonstrates that strategy profiles not corresponding to any pure strategy equilibrium in the base game, such as (Move, Move), cannot arise as pure strategy equilibrium outcomes in the commitment game. Notably, this second result holds only for symmetric 2×2 games, and Section 5 presents a 3×3 game where this is violated.

Lemma 1
Consider a two player normal form game where each player has strategies $A = \{A_1, \ldots, A_n\}$. *Let* $\{A_i, A_{-i}\}$ *be a Nash equilibrium in this game. When this normal form game is played with commitment,*

1. *there is a strategy pair with both players committing to* A_i *and* A_{-i} *respectively, that forms an SPNE of the commitment game.*

2. *if* $U_i(A_i, A_{-i}) \geq U_i(A'_i, B_{-i}(A'_i)) \quad \forall A'_i \in A$, *then there is a strategy pair where* i *commits to* A_i *and* $-i$ *waits and plays* A_{-i} *that forms an SPNE of the commitment game.*

3. *if* $U_i(A_i, A_{-i}) \geq U_i(A'_i, B_{-i}(A'_i)) \quad \forall i \in \{1, 2\}, \forall A'_i \in A$, *then there is a strategy pair where both players wait and then play* A_i *and* A_{-i} *that is an SPNE of the commitment game.*

Proof: I prove each part by showing that the constructed strategy profile yields the outcome (A_i, A_{-i}) and that no unilateral deviation is profitable.

1. Suppose both players commit to A_i and A_{-i}, which gives players payoffs $U(A_i, A_{-i})$. Given the other players commitment, player i can unilaterally only deviate to receive any $U_i(A'_i, A_{-i})$ with $A'_i \in A$. But since (A_i, A_{-i}) is a Nash equilibrium of the normal-form game, no such unilateral deviation can improve i's payoff. By symmetry, this is an equilibrium. To make it subgame perfect, the strategies must also specify equilibrium strategies off-path, for which there are multiple options. All such strategy profiles are subgame perfect.

2. Let player i commit to A_i while player $-i$ waits and plays A_{-i} after observing i's commitment. If i deviates by committing to any $A'_i \in A$ instead, $-i$ will best

respond with $B_{-i}(A_i')$ which yields $U_i\big(A_i', B_{-i}(A_i')\big)$. By assumption this does not exceed $U_i(A_i, A_{-i})$. Similarly, if $-i$ deviates by committing in period 1, the Nash equilibrium property of (A_i, A_{-i}) ensures she cannot gain. Thus no player has a profitable deviation.

3. If both players wait and then, after the commitment period, play A_i and A_{-i}, the outcome is again (A_i, A_{-i}). Any deviation to some commitment A_i' would lead to the opponent responding with $B_{-i}(A_i')$, yielding a payoff $U_i\big(A_i', B_{-i}(A_i')\big)$, which is less than $U_i(A_i, A_{-i})$ by assumption. Thus this is an SPNE: both players get the best equilibrium outcome if they wait, so committing earlier offers no advantage.

Lemma 2

Consider a two player base game, that is a symmetric 2×2 *game with strategies* $\{A_1, A_2\}$ *for each player. If a strategy pair* $\{A_i, A_{-i}\}$ *does not form a Nash equilibrium in the base game, it will not feature in an SPNE of the game with commitment.*

Proof: I prove the contrapositive. If $\{A_i, A_{-i}\}$ features in a commitment game SPNE, it must feature in a base game Nash equilibrium.

1. Let's say players commit to A_i and A_{-i} in the commitment game SPNE. Then, by virtue of this being an SPNE it holds that $U_i(A_i, A_{-i}) \geq u_i(A_i', A_{-i}) \quad \forall i \in \{1, 2\}, \ \forall A_i' \in A$. Thus, $\{A_i, A_{-i}\}$ is a base game Nash equilibrium.

2. Let's say players wait and then play A_i and A_{-i} in the commitment game SPNE. Then, by virtue of this being an SPNE it holds that $U_i(A_i, A_{-i}) \geq U_i(A_i', A_{-i})$ $\forall i \in \{1, 2\}, \ \forall A_i' \in A$. Thus, $\{A_i, A_{-i}\}$ is a base game Nash equilibrium.

3. This is where the 2×2 assumption comes in. Let's say player i commits to A_1 and $-i$ waits. If $-i$ best responds with A_1, by symmetry $\{A_1, A_1\}$ are mutual best responses and thus a Nash equilibrium of the base game. If $-i$ best responds with A_2, one must consider what would have happened if i had committed A_2 instead of A_1. If $-i$'s best response to A_2 would have been A_2, by symmetry i would have preferred to commit A_2 instead of A_1, a contradiction. Thus, $-i's$ best response to A_2 must be A_1, and thus by symmetry $\{A_1, A_2\}$ is a base game Nash equilibrium. By symmetry, this also holds for $\{A_2, A_1\}$. This concludes the proof.

B Proof of Proposition 1

Consider the simplified 2×2 symmetric normal form game from table 2. Let A_1 denote the strategy where a player makes a commitment to A_1, and let A_2 denote the strategy where a player commits to A_2. W denotes the strategy of waiting. To prove Proposition 1, I will derive the mixed subgame perfect equilibria of the

symmetric 2×2 games. In these equilibria, if a player waits, she will best respond to any commitment the other player makes. If both players wait, they will play the MSNE of the base game which yields expected payoff $\frac{bc}{c+b-1}$. Lemma 3 and Lemma 4 (stated and proven further below) ensure that this is the unique mixed subgame perfect equilibrium.

Battle of Sexes ($b > 1$ and $c > 1$) If player $-i$ mixes between A_1, A_2 and W with respective probabilities p, q, and $1-p-q$, then player i's pure strategy payoffs are

$$\begin{aligned} U(A_1) &= p \cdot 0 + q \cdot b + (1-p-q) \cdot b, \\ U(A_2) &= p \cdot c + q \cdot 1 + (1-p-q) \cdot c, \\ U(W) &= p \cdot c + q \cdot b + (1-p-q) \cdot \tfrac{bc}{b+c-1}. \end{aligned}$$

For player i to mix over these strategies, she must be indifferent between them. The p and q that equate the pure strategy payoffs are

$$\begin{aligned} p &= \frac{b(1-b)^2}{(b+c-1)(b^2-b+c^2-c)} \\ q &= \frac{c-(1-p)b}{c-1} \end{aligned}$$

It can be shown that $p, q \in [0,1]$. By symmetry, this holds for player $-i$ as well. To obtain the equilibrium payoff, plug p and q into any of the above pure strategy payoffs. Consider $U(W)$. Note that b and c are both bigger than $\frac{bc}{c+b-1}$. Thus, the payoff in the commitment game equilibrium exceeds the payoff in the base game.

Game of Chicken ($b > 1$ and $1 \geq c > 0$) For $1 \geq c$, waiting weakly dominates A_2. Thus, in the mixed subgame perfect equilibrium there is mixing only over A_1 and W. The probability p that equates $U(A_1)$ and $U(W)$ is

$$p = \frac{b^2-b}{b^2-b+c^2-c+bc}.$$

It can be shown that $p \in [0,1]$. Consider $U(W)$. Since $c \leq 1 \Leftrightarrow c \leq \frac{bc}{c+b-1}$. Thus, the commitment game payoff is worse than the base game payoff for $c < 1$, except for the edge case with $c = 1$ where it is the same.

This concludes the proof of Proposition 1. The next equilibria I derive prove Corollary 1.

Coordination Game ($c < 0$ and $b < 0$) In the Coordination Game and the Stag Hunt the best responses to commitments are different from the best responses in the Game of Chicken and the Battle of Sexes. Here, if player $-i$ mixes over the strategies A_1, A_2, and W with probability p, q, and $1-p-q$, player i's pure strategy payoffs are

$$\begin{aligned} U(A_1) &= p \cdot 0 + q \cdot b + (1-p-q) \cdot 0, \\ U(A_2) &= p \cdot c + q \cdot 1 + (1-p-q) \cdot 1, \\ U(W) &= p \cdot 0 + q \cdot 1 + (1-p-q) \cdot \tfrac{bc}{b+c-1}. \end{aligned}$$

In a mixed equilibrium with mixing over these three strategies, all pure strategy payoffs must be equal. The p and q that equate the above pure strategy payoffs are

$$\begin{aligned} p &= \frac{(1-b)\,(bc-b-c+1)}{(1-b-c)(2bc-b-c+1)}, \\ q &= \frac{p(1-c)-1}{-b}\,. \end{aligned}$$

It can be shown that $p, q \in [0,1]$. To compare the equilibrium payoff of the commitment game to the base game MSNE payoff, plug p and q into $U(W)$. Since $0 > \frac{bc}{c+b-1}$, the commitment game payoff exceeds the MSNE base game payoff.

However, note that committing to (A_1, A_1) or players waiting and then playing (A_1, A_1) also form symmetric equilibria in the commitment game. These equilibria yield payoff 1 for each player, which is bigger than $U(W)$.

Stag Hunt ($0 \leq b < 1$ and $c < 0$) Here, committing to A_1 is weakly dominated by waiting. A_1 will not feature in a mixed equilibrium and p can be set to 0. But then, A_2 weakly dominates waiting. Thus, there is no equilibrium in mixed strategies when commitment is introduced in a Stag Hunt game.

Dominant Strategies When A_1 or A_2 is a (weakly) dominant strategy in the game without commitment, there is little to explore. In equilibrium, players can commit to their dominant strategy, they can wait before playing it or they can mix between committing to it and waiting and then playing it. These equilibria do not offer interesting strategic dynamics, but are mentioned for completeness.

C Proof of Proposition 2

This section proves Proposition 2. I prove 3 lemmas. Lemma 3 states that the proposed mixed equilibrium is the unique subgame perfect equilibrium where players mix over committing and waiting, assuming that players play the base game MSNE if both wait. Lemma 4 shows that when players wait and then play a base game equilibrium in pure strategies, no mixed equilibrium exists in which both players mix over commitment and waiting. Lemma 5 then describes the unique mixed equilibrium as $T \to \infty$ and characterizes closed form expressions for all p_t, q_t and $\sigma(W)$ in terms of b, c and T.

Note that I refer to uniqueness in terms of being the only equilibrium where both players mix over committing and waiting. This definition is more restrictive than necessary. I conjecture that the equilibrium described in Proposition 2 is the unique equilibrium in which miscoordination can occur, which means that both players assign positive probability to committing in some period t. Other equilibria can exist in the game, but they are either in pure strategies or mixed in a trivial sense. For instance, in an equilibrium where Player 1 mixes over committing in period 1 and 2, while Player 2 mixes over committing in period 3 and waiting, Player 1 effectively induces her preferred outcome like in a pure strategy equilibrium.

Lemma 3 (Unique Interior MSNE)
Consider the general symmetric 2×2 *game in Table 2, with parameters* $b > 1$ *and* $c > 0$. *Let each player have the ability to commit to one of two actions,* A_1 *or* A_2, *in any of* T *commitment periods, or to wait until the end. In the subgame where both players wait, they play the mixed-strategy Nash equilibrium of the base game. Then, the commitment game has a unique mixed subgame perfect Nash equilibrium where players mix over committing and waiting. Specifically:*

1. *If* $c > 1$, *both players mix over all strategies* A_{1t}, A_{2t}, *and* W.

2. *If* $c \leq 1$, *both players mix over all strategies* A_{2t} *and* W.

Proof: Remember that $U(A_{1t})$ denotes the payoff of committing to strategy A_1 in period t. p_t and q_t denote the probability of a player playing strategy A_{1t} or A_{2t}, respectively, and $\sigma(W) = 1 - \sum_{\tau=1}^{T}(p_\tau + q_\tau)$ denotes the probability of waiting.

Then, the pure strategy payoffs in a commitment game with T periods are

$$U(A_{1t}) = \sum_{\tau=1}^{t-1} (p_\tau c + q_\tau b) + p_t \cdot 0 + q_t \cdot b + \left(1 - \sum_{\tau=1}^{t} (p_\tau + q_\tau)\right) b \tag{3}$$

$$U(A_{2t}) = \sum_{\tau=1}^{t-1} (p_\tau c + q_\tau b) + p_t c + q_t \cdot 1 + \left(1 - \sum_{\tau=1}^{t} (p_\tau + q_\tau)\right) c \tag{4}$$

$$U(W) = \sum_{\tau=1}^{T} (p_\tau c + q_\tau b) + \sigma(W) \left(\frac{bc}{b + c - 1}\right) \tag{5}$$

To illustrate, suppose player 1 plays A_{1t}. With probability $\sum_{\tau=1}^{t-1}(p_\tau + q_\tau)$, player 2 commits to A_1 or A_2 before her, and player 1 best responds, receiving c or b accordingly. [6] With probability $p_t + q_t$, both players commit simultaneously: if 2 also commits A_1, the payoff is 0; if 2 commits A_2, player 1 receives b. With the remaining probability $1 - \sum_{\tau=1}^{t}(p_\tau + q_\tau)$, player 2 was planning to move only after player 1 committed A_1. In this case, player 1 commits, player 2 best responds with A_2 and 1 receives the payoff b.

For now, consider the case $c > 1$. I am looking for equilibria where players mix over committing and waiting. Thus, $\sigma(W)$ and some p_t or q_t must be greater than 0. First, consider

$$U(A_{1,T}) - U(W) = p_T \cdot (-c) + \sigma(W) \cdot \left(\frac{b(b-1)}{b + c - 1}\right)$$
$$U(A_{2,T}) - U(W) = q_T \cdot (1 - b) + \sigma(W) \cdot \left(\frac{c(c-1)}{b + c - 1}\right)$$

If $p_T = 0$ and $\sigma(W) > 0$, $U(A_{1,T}) > U(W)$. It must be that players play A_{1T} with positive probability, otherwise waiting would be dominated. The same holds for A_{2T}, thus $p_T > 0$ and $q_T > 0$. Next, consider

$$U(A_{1t}) - U(A_{1,t+1}) = p_t \cdot (-c) + p_{t+1} \cdot b$$
$$U(A_{2t}) - U(A_{2,t+1}) = q_t \cdot (1 - b) + q_{t+1} \cdot (c - 1)$$

If $p_t = 0$ and $p_{t+1} > 0$, $U(A_{1t}) > U(A_{1t+1})$. Thus for a player to mix over both A_{1t} and A_{1t+1}, it must be that the other player mixes over both A_{1t} and A_{1t+1} with positive probability. Since both players must have probability $p_T > 0$ as argued above, they must also have probability p_{T-1} greater than 0. It then follows that they must have all probabilities $\{p_t\}_{t=1}^{T}$ greater than 0. The same holds for A_{2t} and q_t. Thus, the

[6] I adopt the convention that $\sum_{\tau=1}^{t-1}(\cdots) = 0$ when $t = 1$, i.e. the sum is empty and thus vanishes.

only equilibrium where both players mix over committing and waiting is such that both players mix over all strategies $\{A_{1t}, A_{2t}, W\}_{t=1}^{T}$ with positive probability. This concludes the proof for $c > 1$.

When $c \leq 1$, playing any A_{2t} is weakly dominated by waiting: equations 4 and 5 imply $U(A_{2t}) \leq U(W)$. Players will not put probability weight on A_2 and thus all $q_t = 0$. The argument of the proof for $c > 1$ then holds for A_1 and W. In equilibrium, players will mix across all strategies in the set $\{A_{1t}, W\}_{t=1}^{T}$. This concludes the proof.

Lemma 4

Consider the general symmetric 2×2 game in Table 2, with parameters $b > 1$ and $c > 0$. Let each player have the ability to commit to one of two actions, A_1 or A_2, in any of T commitment periods, or to wait until the end. In the subgame where both players wait, they play one of the pure strategy Nash equilibria of the base game. Then, as long as $b \neq c$, the commitment game does not have a mixed subgame perfect Nash equilibrium where both players mix over committing and waiting.

Proof: I aim to show that there is no mixed equilibrium in which both players randomize between committing and waiting. In any mixed equilibrium, players must be indifferent between all strategies they assign positive probability to. I will show that no such indifference condition can be satisfied when both player mix between commitment and waiting strategies.

Let each player mix over strategies A_{1t}, A_{2t}, and W with probabilities p_t^i, q_t^i, and $\sigma(W)^i$ for player i, and analogously for player $-i$. Then it holds that

$$U_i(A_{1,t}) - U_i(W) = \sum_{\tau=t+1}^{T} p_\tau^{-i}(b-c) + p_t^{-i}(-c) + \sigma(W)^{-i}\big(b - \max\{b,c\}\big), \tag{6}$$

$$U_i(A_{2,t}) - U_i(W) = \sum_{\tau=t+1}^{T} q_\tau^{-i}(c-b) + q_t^{-i}(1-b) + \sigma(W)^{-i}\big(c - \max\{b,c\}\big), \tag{7}$$

and analogously for player $-i$:

$$U_{-i}(A_{1,t}) - U_{-i}(W) = \sum_{\tau=t+1}^{T} p_\tau^{i}(b-c) + p_t^{i}(-c) + \sigma(W)^{i}\big(b - \min\{b,c\}\big), \tag{8}$$

$$U_{-i}(A_{2,t}) - U_{-i}(W) = \sum_{\tau=t+1}^{T} q_\tau^{i}(c-b) + q_t^{i}(1-b) + \sigma(W)^{i}\big(c - \min\{b,c\}\big). \tag{9}$$

Assume $b > c$. Then from equations (7) and (9), it follows:

$$U_i(A_{2t}) < U_i(W) \quad \text{and} \quad U_{-i}(A_{2t}) \leq U_{-i}(W) \quad \text{for all } t.$$

Thus, neither player will commit to any A_{2t} with positive probability in any mixed equilibrium. The case $b < c$, is symmetric and implies that neither player will commit to A_1. For the remainder of the proof, I focus on the case $b > c$.

Suppose for contradiction that one player mixes over some, but not all, commitment strategies. Then there exists some t such that $p_t = 0$ for one player. Consider the difference in expected utility between adjacent commitment periods:

$$U(A_{1t}) - U(A_{1,t+1}) = p_t \cdot (-c) + p_{t+1} \cdot b \tag{10}$$

Since $p_t = 0$ but $p_{t+1} > 0$, by iterating equation 10 it follows that for the other player $U(A_{1t})$ is bigger than all $U(A_{1t+1}), \ldots, U(A_{1T})$. By equation 6 it follows that $U(A_{1t}) > U(W)$. Therefore, A_{1t} dominates commitment after period t and waiting, contradicting indifference. Thus, for $\sigma(W) > 0$ to hold, both players must assign positive probability to all commitment strategies. However, consider

$$U_i(A_{1,T}) - U_i(W) = p_T^{-i} \cdot (-c) + \sigma(W)^{-i} \cdot (b - \max\{b, c\})$$

Since $b > c$, the second term is zero. If player $-i$ assigns positive probability to A_{1T}, i.e., $p_T^{-i} > 0$, then $U_i(A_{1,T}) < U_i(W)$. This implies that player i cannot be indifferent between waiting and A_{1T} and thus will prefer waiting to committing in any commitment period.

Therefore, it is not possible that both players mix over all commitment strategies and waiting in equilibrium. This concludes the proof.

Lemma 5

Consider the general symmetric 2×2 *game in Table 2 with parameters satisfying* $b > 1$ *and* $c > 0$. *The mixed subgame perfect Nash equilibrium corresponds to the strategy profile described in Table 10. Table 4 summarizes the convergence of equilibrium probabilities and payoffs as* $T \to \infty$.

Proof I will first consider $c > 1$ and then $c \leq 1$.

Case 1: $c > 1$**.** Suppose player $-i$ mixes over A_{1t}, A_{2t} and W with probabilities p_t, q_t and $\sigma(W) = 1 - \sum_{\tau=1}^{T}(p_\tau + q_\tau)$. Then again, player i's pure strategy payoffs are described by equations 3, 4 and 5. In a mixed equilibrium, all these strategies

Strategy	**Expression**	**Case**	
		$c > 1$	$0 < c \leq 1$
Commit A_1 immediately	p_1	$\frac{c-b}{c\ \frac{c-1}{b-1}\left(\frac{c(c-1)}{b(b-1)}\right)^T - b}$	$\frac{1}{\frac{1-\rho^T}{1-\rho}+\rho^T \frac{b+c-1}{b-1}}$
Commit A_2 immediately	q_1	$\frac{p_1 b+c-b}{c-1}$	0
Commit A_1 in period t	p_t	$p_1 \left(\frac{c}{b}\right)^{t-1}$	$p_1 \left(\frac{c}{b}\right)^{t-1}$
Commit A_2 in period t	q_t	$q_1 \left(\frac{b-1}{c-1}\right)^{t-1}$	0
Wait	$\sigma(W)$	$\frac{b+c-1}{b-1} \left(\frac{c}{b}\right)^T p_1$	$\frac{b+c-1}{b-1} \left(\frac{c}{b}\right)^T p_1$

Table 10: Equilibrium Probabilities in the Commitment Game

must yield the same expected payoff:

$$U(A_{11}) = ... = U(A_{1,T}) = U(A_{2,1}) = ... = U(A_{2,T}) = U(W).$$

It can be shown that

$$U(A_{11}) = U(A_{21}) \iff q_1 = \frac{p_1 b + c - b}{c - 1} \tag{11}$$

$$U(A_{1t}) = U(A_{1t+1}) \iff p_{t+1} = p_t \cdot \frac{c}{b} \tag{12}$$

$$U(A_{2t}) = U(A_{2t+1}) \iff q_{t+1} = q_t \cdot \frac{b-1}{c-1} \tag{13}$$

$$U(A_{1T}) = U(W) \iff \sigma(W) = p_T \cdot \frac{c}{b} \cdot \frac{b+c-1}{b-1} \tag{14}$$

$$U(A_{2T}) = U(W) \iff \sigma(W) = q_T \cdot \frac{b-1}{c-1} \frac{b+c-1}{c} \tag{15}$$

I now want to express p_t, q_t and $\sigma(W)$ as closed form solutions in terms of b, c and T. First note that 12 and 13 can be used to write

$$p_t = p_1 \left(\frac{c}{b}\right)^{t-1} \qquad q_t = q_1 \left(\frac{b-1}{c-1}\right)^{t-1}$$

Next, I will use these to plug into 14 and 15 to obtain

$$p_1 \left(\frac{c}{b}\right)^T \frac{b+c-1}{b-1} = q_1 \frac{b+c-1}{c} \left(\frac{b-1}{c-1}\right)^T$$

Next, I plug in 11 for q_1 to obtain an expression that can be rearranged for p_1. This

finally yields:

$$p_1 = \frac{c-b}{c\frac{c-1}{b-1}\left(\frac{c(c-1)}{b(b-1)}\right)^T - b}, \tag{16}$$

I have thus derived a closed form expression for p_1, and using equations 11-15 I can obtain closed form expressions for all p_t, q_t and $\sigma(W)$. Next, let's analyze the equilibrium as $T \to \infty$. Define $r = \frac{c(c-1)}{b(b-1)}$. Then

$$p_1 = \frac{c-b}{c\frac{c-1}{b-1}r^T - b}, \quad q_1 = \frac{p_1 b + c - b}{c-1} \quad \text{and} \quad \sigma(W) = p_1 \left(\frac{c}{b}\right)^T \frac{b+c-1}{b-1}.$$

As $T \to \infty$, I distinguish three cases:

- If $c < b$, then $r < 1$, and

$$p_1 \to \frac{b-c}{b}, \qquad q_1 \to 0, \qquad \sigma(W) \to 0.$$

- If $c > b$, then $r > 1$, and

$$p_1 \to 0, \qquad q_1 \to \frac{c-b}{c-1}, \qquad \sigma(W) \to 0.$$

- Finally, if $c = b$

$$p_1 = \frac{1}{T+1}\frac{b-1}{2b-1}, \quad q_1 = \frac{1}{T+1}\frac{b}{2b-1}, \quad \sigma(W) \to 0$$

Note that the case $b = c$ is not derived from equation 16. Instead, I plug equations 11-15 into $\sum_{t=1}^{T}(p_t + q_t) + \sigma(W) = 1$ to obtain closed form expressions. To see the equilibrium payoff in all cases consider $U(W)$. As can be seen, it converges to $\min\{b, c\}$ as $T \to \infty$.

Case 2: $c \leq 1$**.** Here, committing to A_2 is weakly dominated by waiting. Thus all $q_t = 0$ and the mixed equilibrium is only characterized by Equation 12 and 14. Let $\rho = \frac{c}{b}$. Since the probability sequence must satisfy $\sum_{\tau=1}^{T} p_\tau + \sigma(W) = 1$, I can write

$$p_1 \sum_{\tau=0}^{T-1} \rho^\tau + p_1 \rho^T \frac{b+c-1}{b-1} = 1.$$

The sum of the geometric series is $\sum_{\tau=0}^{T-1} \rho^\tau = \frac{1-\rho^T}{1-\rho}$. Plugging this into the above

expression and solving for p_1 gives

$$p_1 = \frac{1}{\frac{1-\rho^T}{1-\rho} + \rho^T \frac{b+c-1}{b-1}},$$

which together with 12 and 14 give the closed form expressions for all p_t and $\sigma(W)$. Again, let me analyze the equilibrium as $T \to \infty$. Note that $\rho^T \to 0$, so that the geometric series simplifies to $\frac{1-\rho^T}{1-\rho} \to \frac{1}{1-\rho}$. Thus,

$$p_1 \to 1 - \rho, \qquad \sigma(W) \to 0.$$

Consider $U(W)$ again to see that the equilibrium payoff converges to c. Thus fur $c < 1$ the equilibrium payoff worsens as T increases. Note that $c = 1$ is an edge case, where the MSNE base game payoff $\frac{bc}{b+c-1} = 1$ and thus the commitment game equilibrium payoff stays 1 as $T \to \infty$. This concludes the proof of Lemma 5.